

Yukawova interakce

→ simplifikovaný model pro pionové interakce s nukleony

- interakce Diracovského pole a reálného skalárního pole

ϕ ... pseudoskalární pole

- $s=0$
- mass M
- $\eta_P = -1$

ψ ... Diracovské pole

- $s=1/2$
- mass m
- η_P jakéhokoliv

→ požadavky na $\mathcal{L}_I$

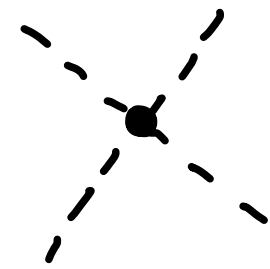
- Lorentz invariantní → skalár
- lokální
- invariance vůči paritě
- invariance vůči nábojovému sdružení
- hermitita
- renormalizabilita $[\phi] \leq 4$
- $U(1)$ symetrie

$$\Rightarrow \mathcal{L}_I \ni \phi \bar{\psi} \Gamma \psi \quad \Gamma \in \left\{ \begin{array}{l} 1 \text{ skalár} \\ \gamma_5 \text{ pseudoskalár} \end{array} \right.$$

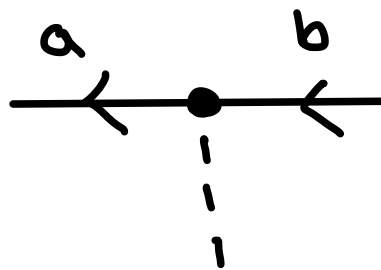
$$\Rightarrow \mathcal{L}_I = \underbrace{-ig \bar{\psi} \gamma_5 \psi \phi}_{\text{hermitita}} + \frac{\lambda^4}{4!} \phi^4$$

Feynmanovy pravidla

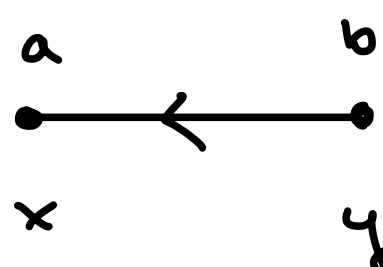
x-representace



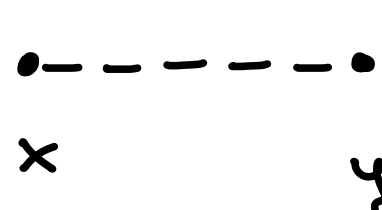
$$\equiv i\lambda$$



$$\equiv ig\gamma_5^{ab}$$

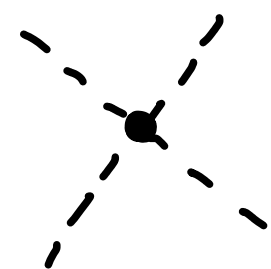


$$\equiv iS_F(x-y)$$

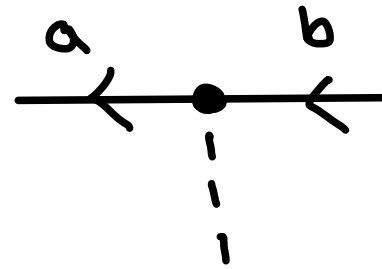


$$\equiv i\Delta_F(x-y)$$

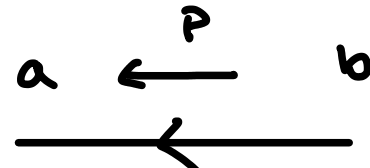
p-representace



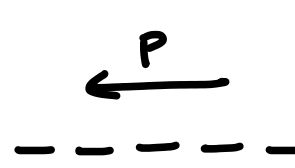
$$\equiv i\lambda(2\pi)^4\delta^4(p)$$



$$\equiv g\gamma_5^{ab}(2\pi)^4\delta^4(p)$$



$$\equiv \frac{i}{\not{p} - m + i0} \quad ab$$

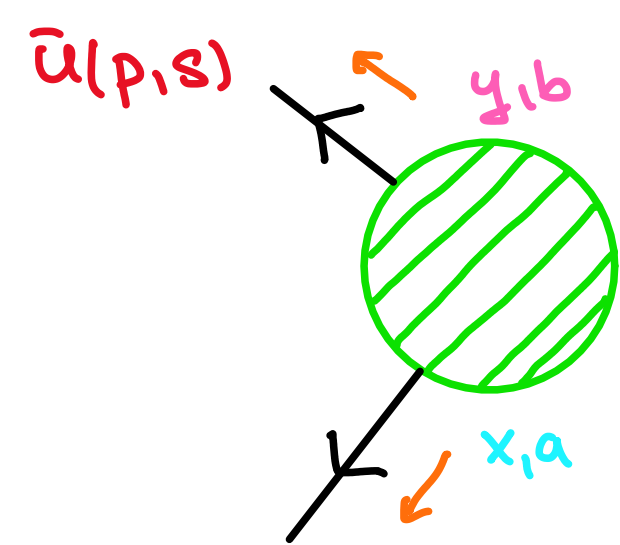


$$\equiv \frac{i}{p^2 - M^2 + i0}$$

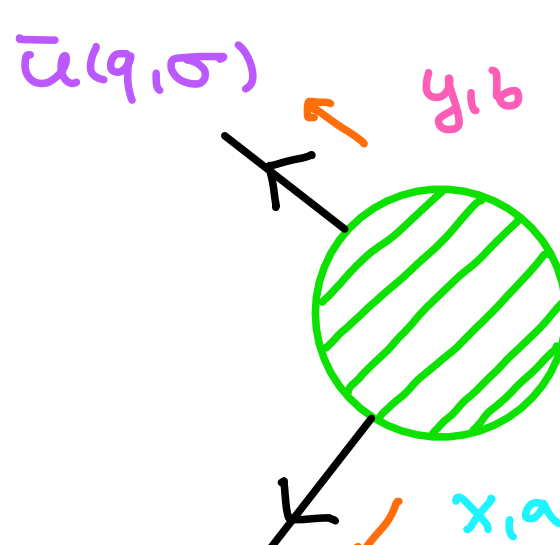
S-matice

→ kombinatorické úvahy se nemění $S = 2^B \prod_{\langle v, v' \rangle} n_{vv'}! g$

→ topologicky ekvivalentní grafy s rozdílným počtem vnějších fermionových linek se liší relativním znaménkem



$$W_I(p, q)$$



$$W_{II}(q, p)$$

$$W_I(p, q) = -W_{II}(q, p) \quad \left| \quad \bar{u}(p, s) \leftrightarrow \bar{u}(q, \sigma) \right.$$

$$\text{protože} \quad : b(p, s) b(q, \sigma) X \bar{\psi}(x) Y \bar{\psi}(y) Z :$$

výše uvedených dvojic existuje $\ominus$

Yukavův potenciál

$$\rightarrow \mathcal{T}_{fi}^{\text{Born}} = - \langle \vec{p}, s, \bar{s} | V | \vec{k}, \sigma, \bar{\sigma} \rangle$$

$$\Rightarrow V(\vec{x}) = \int e^{-i\vec{q} \cdot \vec{x}} \mathcal{T}_{fi}^{\text{Born}}(\vec{q}) d\vec{q}$$

$$\rightarrow \mathcal{T}_{fi}^{(+)} = -g^2 \frac{1}{p^2 - M^2} \underbrace{\chi_s^\dagger \vec{\sigma} \cdot \vec{q} \chi_\sigma}_{\text{spinová interakce}} \underbrace{\chi_{\bar{s}}^\dagger \vec{\sigma} \cdot \vec{q} \chi_{\bar{\sigma}}}$$

$$\Rightarrow V(\vec{x}) = 2\pi^2 \frac{e^{-Mx}}{x} \quad \text{Yukawa}$$